

MCA SEMESTER-I
School of Computer and Systems Sciences,
Jawaharlal Nehru University,
New Delhi -110067
Programming in C (CS-401)

Mid Exam

Total Marks:20

Date: 21.11.2025

Time: 2hr

All Compulsory

1. In C, Write note on:
 - a. Difference between = and ==
 - b. Float and double in C
 - c. break and continue

3*2=6
2. Discuss High level and low level languages, assembly language with examples. 4
3. Discuss flow chart and C program to print all numbers from 1 to 20 with the increment of 2. 5
4. Discuss Pseudo code and C program that reads the marks of a student and prints:
Marks are user input.
"Distinction" if marks ≥ 75
"Pass" if marks ≥ 40 and < 75
"Fail" otherwise 5

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End semester Exam

Total Marks:50

Date: 12.12.2025

Time: 3hr

1. Write note on:
 - a. Struct and union
 - b. Preprocessor directives
 - c. Keywords in C
 - d. Malloc and calloc
 - e. Address operator with examples4x5=20
2. Program to print the sum of digits of a user input number using function. 6
3. Write a program to check if the number elements is even or odd using functions 6
4. Write a program to check whether a person is eligible to cast a vote. Take input as year of birth from the user. Also, check for a negative age. 6
5. Write a program to calculate the electricity bill. Given units: 6
 - first 100 units → ₹10/unit
 - next 100 → ₹14/unit
 - above 200 → ₹20/unit

Calculate the total amount using a function.
6. Write a C program to insert a number entered by the user into an array at a user-defined position. The program should:
 - Take the size of the array and its elements as input from the user.
 - Ask the user for the number to be inserted.
 - Ask for the position where the number should be inserted.
 - Shift the elements accordingly and display the updated array.

OR

Write a program to create two user-defined arrays 1 and 2, take size and elements of array from user. Merge the array 1 and 2 into a third array and print the third array. 6